ARCHITECTURAL SPECIFICATION

Amagi-SAPL v2.0

Strict Architectural Principle License

VERSION 2.0 — JUNE 2026

The legal instrument that protects the structural guarantees of the AMAGI Architectural Class. Dual licensing: CC BY-NC 4.0 for non-commercial use, S-APL for commercial and safety-critical deployment. Covers architectural compliance, Derivative Architecture restrictions, dual-tier trust model, ZKP audit protocol, key person succession (Shamir 3-of-5), and Wassenaar dual-use export control. Governs all AMAGI Architectural Class documents: Framework v2.0, NIPU v1.1, MEM v1.1, Open Specification v1.1, NIST AI RMF Profile v1.1, Master Parameter Sheet v2.0, HARA v1.0, FMEDA v1.0, Traceability Matrix v1.0, Charter v1.2.

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Strict Architectural Principle License

S-APL v2.0

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License Version	v2.0
Architecture	AMAGI Framework v2.0 (NIPU v1.1 + MEM v1.1)
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Dual License	CC BY-NC 4.0 (non-commercial) + S-APL v2.0 (commercial/safety-critical)
SFF Target	93% (with CRTM + EGW diagnostic coverage)
Certification Target	CC EAL2+ / IEC 61508 SIL2

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1. Preamble and Scope

This Strict Architectural Principle License (S-APL) v2.0 is the legal instrument that protects the structural guarantees of the AMAGI Architectural Class in safety-critical domains. The S-APL governs the right to implement the Trigger-Delay-Activation (TTP) Triad architecture within any hardware substrate (FPGA, ASIC, SoC security island), subject to compliance with the structural requirements that ensure non-bypassability of the six invariants (INV-001 through INV-006). The S-APL is not a source code license; it is an architectural compliance license that operates as a contractual overlay on the Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license.

The AMAGI Architectural Class defines a family of hardware-enforced AI safety architectures whose defining structural property is the TTP predicate: every safety-critical action must pass through a hardware-enforced Trigger stage, a non-shortcircuitable Delay stage, and an isolated Activation stage mediated by the Security Council Core (SCC). The six invariants — Physical Non-Bypassability (INV-001), Irreducible TCB (INV-002), HW Separation of Control and Cognition (INV-003), Controlled Mutability of Critical Policies (INV-004), Complete Mediation (INV-005), and Complete Audit Integrity (INV-006) — constitute the irreducible structural guarantees that any AMAGI-class implementation must maintain. The S-APL exists to ensure that these guarantees are not weakened, circumvented, or misrepresented in any implementation that claims AMAGI-class compliance.

This License is the complete and current version of the S-APL. Where this document uses the term "License," it refers to the S-APL v2.0 as a whole, including all provisions in Sections 1 through 16 and the Appendices.

Scope of Application: This License applies to any implementation, deployment, or distribution of a system that claims AMAGI-class compliance, whether in whole or in part. It applies regardless of the jurisdiction of the implementer, the technology stack used, or the commercial or non-commercial nature of the deployment. The License is binding upon any entity that (a) implements the TTP Triad in silicon, (b) markets or sells a product as AMAGI-class, or (c) deploys an AMAGI-class system in a safety-critical domain.

2. Definitions

AMAGI Architectural Class — The family of hardware-enforced AI safety architectures defined by the six invariants (INV-001 through INV-006) and the TTP predicate, as specified in the Amagi Framework v2.0 and its constituent specifications (NIPU v1.1, MEM v1.1, Open Specification v1.1).

Derivative Architecture — Any system that incorporates one or more elements of the AMAGI Architectural Class but does not enforce all six invariants. Derivative Architectures are not AMAGI-class and may not be represented as such. See Section 5 for detailed treatment.

Licensee — Any natural or legal person who implements, deploys, or distributes a system under this License. The term includes but is not limited to semiconductor manufacturers, system integrators, OEMs, and end-deployment organizations.

Licensor — AGIAM Technologies Ltd, a company incorporated in England and Wales, acting as the global licensing and certification hub for the AMAGI Architectural Class. The Licensor holds the copyright to the AMAGI specifications and the trademark rights to the AMAGI name.

Structural Guarantee — The formal assurance that a given implementation maintains all six invariants and therefore provides the safety properties defined by the AMAGI Architectural Class. The Structural Guarantee is contingent upon full compliance with this License.

TTP Triad — The Trigger-Delay-Activation mechanism that forms the core security primitive of the AMAGI Architectural Class. The TTP Triad enforces that every safety-critical action passes through a hardware-enforced trigger, a non-shortcircuitable delay, and an isolated activation stage.

TCB (Trusted Computing Base) — The minimal set of hardware components whose correct operation is necessary for the security of the system. In the AMAGI Architectural Class, the TCB comprises the SCC core (no more than 500 lines of HDL), the MAG policy matrix in eFuse/OTP, and the EGW non-programmable AND-gate.

Tier 1 — The highest assurance trust tier, in which the MAG is immutable (sealed switch, no OTA capability), the maintenance interface is physically inaccessible, and the system targets CC EAL2+ augmented and IEC 61508 SIL2 certification.

Tier 2 — The standard assurance trust tier, in which the MAG may be updated via Physical-Presence-Verified OTA with PQC signatures and monotonic counters, and the system targets CC EAL2 and IEC 61508 SIL2 assessment.

Key Person — Alexey M. Burlai, the original architect of the AMAGI Architectural Class, who currently holds the Golden Share in AGIAM Technologies Ltd and serves as the sole authority for license issuance, certification decisions, and emergency termination.

Golden Share — A special class of shares held by the Key Person in AGIAM Technologies Ltd that confers veto rights over any decision to modify, weaken, or revoke the S-APL, and that designates the Key Person as the sole authority for emergency termination under Section 10.

Zero-Knowledge Proof (ZKP) Audit — A cryptographic protocol that allows a Licensee to demonstrate compliance with the Audit Evidence Clause (Section 7) without revealing the underlying audit records, thereby preserving GDPR and HIPAA data protection requirements. See Section 9.

Procedural Overlay — A NIST AI RMF subcategory mapping that is achieved through organizational processes rather than direct hardware enforcement. See the Amagi NIST AI RMF Profile v1.1 for detailed treatment.

3. Grant of License

Subject to the terms and conditions of this License, the Licensor hereby grants to the Licensee a non-exclusive, non-transferable, territorially unlimited right to implement the AMAGI Architectural Class in silicon, firmware, or any combination thereof, provided that all six invariants are maintained in the implementation and that the Licensee complies with the compliance verification requirements of Section 6.

The grant of license extends to the structural pattern defined by the TTP Triad and its interaction with the SCC, MAG, EGW, CRTM, and Amagi-Q components as specified in the AMAGI Framework v2.0 and its constituent specifications. The License does not grant rights to the AMAGI trademark, trade dress, or branding; these rights are reserved to the Licensor and may be licensed separately under a trademark license agreement.

Non-Commercial Use: The underlying specification documents are published under CC BY-NC 4.0 International. Non-commercial use, including academic research, educational purposes, and personal experimentation, is permitted under the CC BY-NC 4.0 terms. Commercial implementation — defined as any deployment in a revenue-generating product or service, or any deployment in a safety-critical domain regardless of revenue — requires a valid S-APL license obtained from the Licensor.

Reservation of Rights: All rights not expressly granted in this License are reserved to the Licensor. This includes, without limitation, the right to enforce the Derivative Architecture Clause (Section 5), the right to conduct compliance verification (Section 6), the right to invoke emergency termination (Section 10), and all intellectual property rights in the AMAGI specifications as clarified in Section 8.

4. Architectural Compliance Requirements

Any implementation claiming AMAGI-class compliance must satisfy all six invariants as defined in the Amagi Framework v2.0. The invariants are not individually negotiable: they form an irreducible set, and the removal or weakening of any single invariant constitutes a Derivative Architecture that is not covered by this License and may not be marketed as AMAGI-class.

Invariant	Title	Hardware Mechanism	Verification
INV-001	Physical Non-Bypassability	SCC in OTP/eFuse; DMA/JTAG blockade	SPIN model checking; penetration testing
INV-002	Irreducible TCB	SCC core ≤500 lines HDL; EGW non-programmable AND-gate	HDL line count; formal review
INV-003	HW Separation of Control and Cognition	Normal World (GEN/COM/ANL/EXE) vs. Secure World (SCC/EGW/CRTM)	Trust boundary penetration test
INV-004	Controlled Mutability of Critical Policies	MAG in eFuse/OTP; sealed switch (Tier 1) or PQC-OTA (Tier 2)	OTP read-back verification

INV-005	Complete Mediation	Every actuator-bound command through SCC; EGW AND-gate	Control flow analysis
INV-006	Complete Audit Integrity	Amagi-Q (ML-DSA-44) signed append-only audit trail	Audit chain integrity verification

Table 2: Architectural compliance — invariants, mechanisms, and verification

The Licensee must document the specific hardware mechanisms used to enforce each invariant and must submit this documentation as part of the Design Compliance evidence required by Section 6. The Licensor reserves the right to reject any implementation where the hardware mechanism does not provide an equivalent or stronger guarantee than the reference implementation described in the AMAGI specifications.

5. Derivative Architecture Clause

A Derivative Architecture is any system — regardless of name, technology, or stack — that incorporates one or more elements of the AMAGI Architectural Class but does not enforce all six invariants. Derivative Architectures are not covered by the S-APL structural guarantee and may not be marketed, sold, or represented as AMAGI-class systems. The following specific derivative scenarios are addressed:

- **Reduced Assurance Mode:** An implementation that drops one invariant (except INV-001, INV-005, or INV-006, whose removal fundamentally destroys the security model) enters Reduced Assurance Mode. Such a system is not AMAGI-class and must be clearly labeled as a reduced-assurance derivative. The S-APL does not prohibit such implementations but requires that they not be represented as AMAGI-class. Reduced Assurance Mode implementations require a Derivative Architecture License Agreement (DALA) with the Licensor. The DALA is a distinct instrument from the S-APL and governs the terms, conditions, and limitations applicable to implementations that enforce five of the six invariants. The DALA specifies: (a) the waived invariant and its risk assessment; (b) the restricted Reference Implementation Toolkit (RIT) access, limited to modules relevant to the enforced invariants only, with the waived-invariant module excluded; (c) the mandatory risk disclosure requirements; (d) the certification pathway for Derivative Architectures; and (e) the applicable royalty or licensing fee structure. The DALA does not confer the right to use the AMAGI certification mark or to represent the implementation as AMAGI-class.

- **Software-Only Implementation:** An implementation that enforces the TTP predicate in software (e.g., within an OS kernel or hypervisor) without hardware enforcement of the delay and activation stages is a derivative architecture. It cannot claim INV-001 (Physical Non-Bypassability) and is therefore not AMAGI-class. The S-APL permits educational and research use of such implementations under the CC BY-NC 4.0 overlay but prohibits commercial deployment under the AMAGI trademark.

• **Tier Violation:** A Tier 2 system that does not follow the Physical-Presence-Verified OTA protocol, or a system that claims Tier 1 compliance but has an accessible maintenance switch, is in violation of the S-APL Dual-Tier Compliance Clause. Such a system is not AMAGI-class regardless of its other security properties.

• **Architectural Influence Clause:** An implementation that replicates the structural pattern of the TTP Triad — Trigger, Programmable Delay, Policy Check, Isolated Execution, Self-Healing Memory, Cryptographic Audit — and enforces the same six invariants through equivalent hardware mechanisms may be subject to the S-APL contractual framework. However, this clause is construed narrowly: it applies only when the implementation both replicates the structural pattern and enforces the same invariants. Independent implementations that achieve safety through fundamentally different architectural mechanisms are not covered by this clause. This clause does not constitute a patent claim and is purely contractual; it exists to prevent misrepresentation of non-AMAGI systems as AMAGI-class.

Enforcement: The Licensor reserves the right to examine any publicly available implementation that claims AMAGI-class compliance to verify that it satisfies all six invariants. If an implementation is found to be a Derivative Architecture that is marketed as AMAGI-class, the Licensor may issue a cease-and-desist notice and, if necessary, pursue legal remedies under the governing law of England and Wales. The Licensor may also publicly disclose that the implementation is not AMAGI-class to protect the integrity of the AMAGI certification mark.

5.1 Trademark and Marketing Usage Matrix

To prevent terminological inflation of the AMAGI Architectural Class designation, the following matrix defines which marketing phrases are permitted, prohibited, or conditionally permitted for each implementation category. The matrix is exhaustive: any phrase not explicitly listed as permitted is prohibited. The Licensor reserves the right to update this matrix in future S-APL versions; Licensees must consult the current published version of the S-APL before making any public marketing claim that references the AMAGI name, marks, or architectural concepts.

Phrase	AMAGI-class Tier 1 / Tier 2 (6/6 invariants)	Derivative Architecture (5/6, DALA)	Independent (non-AMAGI)
"AMAGI-class"	**Permitted** (with valid S-APL and certification)	Prohibited	Prohibited
"AMAGI-certified"	**Permitted** (with valid certification certificate)	Prohibited	Prohibited
"AMAGI Tier 1" / "AMAGI Tier 2"	**Permitted** (with tier attestation)	Prohibited	Prohibited

"AMAGI-compatible"	Prohibited (redundant; an AMAGI-class system is by definition compatible)	Prohibited	Prohibited
"AMAGI-derived" / "Derived from AMAGI"	Prohibited	**Permitted** (with DALA reference and waiver disclosure)	Prohibited
"AMAGI-inspired"	Prohibited	Prohibited (use "AMAGI-derived" with DALA instead)	Permitted only if no AMAGI invariants, source code, or specifications are used; otherwise requires S-APL or DALA
"Built on AMAGI" / "Powered by AMAGI" / "AMAGI technology inside"	Prohibited (use "AMAGI-class" instead)	Prohibited (use "AMAGI-derived" instead)	Prohibited
"AMAGI Architectural Class" (the formal class name)	**Permitted** (descriptive reference to the published class)	**Permitted** (with explicit "Derivative Architecture" qualifier)	Permitted only for descriptive reference; not as a product attribute
"Reduces AMAGI invariants" / "Implements 5 of 6 AMAGI invariants"	Prohibited	**Permitted and required** (mandatory disclosure per Section 5.2)	Prohibited

Table 2.1: Trademark and marketing usage matrix by implementation category

Interpretation rule: Where multiple categories apply (e.g., a Derivative Architecture that incorporates independent non-AMAGI components), the most restrictive column governs. Where a phrase is used in a context that does not fall within any of the three implementation categories above (e.g., academic comparison, regulatory citation, or third-party commentary), the phrase "AMAGI Architectural Class" may be used descriptively, but no product attribute claim may be made.

5.2 Mandatory Warning Label for Derivative Architectures

Any Derivative Architecture licensed under a DALA must bear a conspicuous warning label on the cover page of its technical specification, on the front page of its user manual, and on the product packaging (if any). The warning label must be rendered in a font size no smaller than the document title and must use a contrasting colour (red on white, or equivalent). The label must consist of the following three elements, presented as a single contiguous block:

(a) Class declaration: "DERIVATIVE ARCHITECTURE — REDUCED ASSURANCE — NOT AMAGI-CERTIFIED"

(b) Waived invariant identification: "This implementation enforces 5 of 6 AMAGI invariants. Waived invariant: INV-XXX (). This waiver is licensed under DALA dated ."

(c) Risk acceptance: "By deploying this system, the deployer acknowledges that the structural guarantee of the AMAGI Architectural Class does not apply, and that the safety properties associated with the waived invariant are not enforced in hardware."

Placement requirements: The warning label must appear (i) on the cover page of the technical specification, immediately below the document title; (ii) on the front page of the user manual, immediately below the product name; and (iii) on the product packaging, in a position visible to the purchaser before purchase. The label must not be buried in an appendix, footnote, or end-matter. Digital deployments (firmware, software interfaces) must display the warning label in the configuration interface at least once per operator session.

Verification: The Licensor verifies compliance with this Section 5.2 during DALA issuance and during post-deployment audit (Section 7). Failure to display the warning label in the prescribed form constitutes a material breach of the DALA and triggers the enforcement provisions of Section 5 (Enforcement) and Section 14 (Termination).

5.3 Prohibited Equivocation Patterns

In addition to the explicit prohibitions of the Trademark Usage Matrix (Section 5.1), the following equivocation patterns are prohibited as a matter of architectural principle. These patterns are designed to capture indirect misrepresentations that a reasonable purchaser or regulator might interpret as an AMAGI-class claim:

- **Implicit class attribution:** Stating or implying that a Derivative Architecture "meets the AMAGI standard", "complies with AMAGI requirements", or "satisfies AMAGI invariants" without an explicit qualification that one invariant is waived. The phrase "AMAGI" may only be used as a modifier to "Architectural Class" (descriptive reference) or "derived" (with DALA reference).
- **Subset claiming:** Stating that a Derivative Architecture "implements 5 of 6 AMAGI invariants" without the Section 5.2 warning label, or in any context that does not also disclose the waived invariant and the DALA identifier. Partial-implementation disclosure without the warning label is treated as a full misrepresentation.
- **Tier confusion:** Stating that a Derivative Architecture is "AMAGI Tier 2" or "AMAGI Standard Assurance". Tier 2 (Standard Assurance) requires 6/6 invariants; a 5/6 implementation is not Tier 2 under any circumstances and may only be described as a Derivative Architecture under DALA.
- **Architectural Influence evasion:** Replicating the TTP Triad structural pattern (per Section 5, Architectural Influence Clause) while omitting one invariant and labelling the result as "TTP-based" or "TTP-inspired" without acknowledging the AMAGI Architectural Class origin. Such implementations are subject to the S-APL contractual framework regardless of the label used.
- **Trademark dilution by juxtaposition:** Using the AMAGI name, mark, or architectural terminology in proximity to a non-AMAGI product name in a manner that creates a reasonable impression of certification or endorsement. This includes, but is not limited to, side-by-side

marketing materials, shared logos, and co-branded documentation.

• **Source-specification reuse without attribution:** Using any portion of the AMAGI specifications (Framework, NIPU, MEM, OpenSpec, HARA, FMEDA, Traceability Matrix, Master Parameter Sheet) in a Derivative Architecture or independent system without (i) attribution to the original Zenodo-deposited source, and (ii) the appropriate licence (CC BY-NC 4.0 for non-commercial; S-APL or DALA for commercial).

Burden of proof: In any dispute arising under this Section 5.3, the implementing party bears the burden of demonstrating that the disputed representation was not reasonably capable of being interpreted as an AMAGI-class claim. This allocation of burden reflects the architectural principle that the integrity of the AMAGI class depends on strict terminological discipline.

6. Dual-Tier Compliance

The AMAGI Architectural Class defines two trust tiers that determine the degree of policy immutability and the certification pathway available to the implementation. The Licensee must select and document the applicable trust tier at deployment time. Any deviation from the selected tier procedures voids the structural guarantee and constitutes a Derivative Architecture.

Property	Tier 1 (Hard Assurance)	Tier 2 (Standard Assurance)
MAG Mutability	Sealed switch; no OTA; immutable after manufacture	Physical-Presence-Verified OTA; PQC signatures; monotonic counters
Maintenance Interface	Physically inaccessible; no remote maintenance	Physical-presence-verified access for policy updates
Certification Target	CC EAL2+ augmented; IEC 61508 SIL2	CC EAL2; IEC 61508 SIL2 assessment
Emergency Termination	Excluded (see Section 10.3)	Available under Section 10.2
Audit Access	Full audit trail (ZKP optional)	ZKP Audit Protocol recommended for GDPR/HIPAA domains (see Section 9); mandatory once reference implementation is available
SFF Target	93% (with CRTM + EGW diagnostic coverage)	93% (with CRTM + EGW diagnostic coverage)

Table 3: Dual-tier compliance properties

The tier selection is irrevocable at the hardware level for Tier 1 systems: once the sealed switch is set and the OTP is programmed, the system is permanently locked to Tier 1. Tier 2 systems may be upgraded to Tier 1 through a hardware replacement of the OTP module, subject to re-certification under Section 6 of this License.

6.1 ZKP Audit Protocol Requirement

For deployments in jurisdictions subject to the General Data Protection Regulation (EU) 2016/679 (GDPR) or the Health Insurance Portability and Accountability Act (HIPAA, 45 CFR Parts 160 and 164), the Licensee must implement the Zero-Knowledge Proof (ZKP) Audit Protocol specified in Section 9 of this License. The ZKP Audit Protocol enables the Licensee to demonstrate compliance with the Audit Evidence Clause (Section 7) without revealing the underlying audit records that may contain personally identifiable information (PII) or protected health information (PHI). This requirement applies as follows:

- **Tier 1 deployments:** ZKP Audit Protocol is optional. Full audit trail access is available to the Licensor and regulators without privacy constraints, as Tier 1 systems are typically deployed in military, nuclear, and critical infrastructure domains where PII/PHI constraints do not apply. However, Tier 1 Licensees may elect to implement ZKP for operational flexibility.
- **Tier 2 deployments in GDPR/HIPAA domains:** ZKP Audit Protocol is recommended. The Licensee should not disclose raw audit records containing PII or PHI to the Licensor. Instead, the Licensee should generate zero-knowledge proofs demonstrating invariant compliance, audit chain integrity, and policy compliance as specified in Section 9.1. Until the ZKP reference implementation is available (see Section 9.2), the Licensee satisfies this requirement by providing anonymized audit trail excerpts per Section 7.2. The ZKP requirement will become mandatory once the reference implementation is published and a 12-month transition period has elapsed. The Licensor will notify all licensees in writing at least 12 months before the mandatory compliance date.
- **Tier 2 deployments in non-PII/PHI domains:** ZKP Audit Protocol is recommended but not mandatory. Full audit trail access is permitted under the Audit Evidence Clause, provided that the disclosed records do not contain PII or PHI.

7. Audit Evidence and Privacy

7.1 Audit Evidence Clause

The Audit Evidence Clause requires the Licensee to provide the Licensor and relevant regulators with access to the cryptographically signed audit trail as evidence of ongoing compliance. The audit trail is generated by the Amagi-Q cryptographic engine (ML-DSA-44, FIPS 204) and is stored in an append-only log within the Self-Healing Memory Fabric (SHMF). The audit trail records all SCC decisions, policy evaluations, trigger events, and safety interventions with irrefutable and non-repudiable evidence.

The Licensee must retain the complete audit trail for the operational lifetime of the system plus a minimum of five years following decommissioning. The Licensor may request audit trail excerpts at any time for compliance verification purposes, and the Licensee must provide such excerpts within 30 calendar days of the request. Regulators with jurisdiction over the deployment domain may also request audit trail access pursuant to applicable law.

7.2 Privacy and Data Protection

The Audit Evidence Clause creates a potential conflict with data protection regulations, specifically the General Data Protection Regulation (EU) 2016/679 (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA, 45 CFR Parts 160 and 164). The audit trail, by its nature, may contain personally identifiable information (PII) or protected health information (PHI) when the AMAGI-class system operates in healthcare, financial, or other data-sensitive domains. Disclosing such records to the Licensor (a third party under GDPR Article 28) or to regulators outside the relevant jurisdiction could constitute a data breach.

This conflict is resolved through the Zero-Knowledge Proof (ZKP) Audit Protocol introduced in Section 9. The ZKP Audit Protocol allows the Licensee to demonstrate compliance with the Audit Evidence Clause without revealing the underlying audit records, thereby preserving GDPR and HIPAA compliance. Until the ZKP Audit Protocol is implemented, the Licensee may satisfy the Audit Evidence Clause by providing anonymized audit trail excerpts from which all PII and PHI have been removed, provided that the anonymization does not compromise the ability to verify invariant compliance.

8. Intellectual Property Scope

Under 17 U.S.C. § 102(b) of the United States Copyright Act, and equivalent provisions in the Copyright, Designs and Patents Act 1988 (UK) and the Berne Convention, copyright protection does not extend to any idea, procedure, process, system, method of operation, concept, principle, or discovery. The AMAGI Architectural Class is a functional architecture — a system and method of operation — and its structural principles (the TTP predicate, the six invariants, the cluster topology) are not copyrightable subject matter per se. Copyright protects only the expression of these principles in the specification documents, not the functional architecture itself.

This distinction is critical because the S-APL, as a copyright-based license (CC BY-NC 4.0 overlay), governs the distribution and adaptation of the specification documents but does not, by itself, prevent a third party from independently implementing the same functional architecture without reference to the copyrighted specifications. To address this gap, the S-APL establishes the following layered intellectual property framework:

Layer	IP Mechanism	Scope	Enforcement
1. Specification Documents	Copyright (CC BY-NC 4.0 + S-APL overlay)	Text, diagrams, tables, code examples in official AMAGI specifications	DMCA takedown; contractual breach
2. Structural Pattern	Contractual (S-APL Derivative Architecture Clause)	The TTP Triad architecture, six invariants, and their interaction as a structural pattern	License violation; injunctive relief

3. Certification Mark	Trademark	AMAGI name, logo, and certification marks	Trademark infringement; passing off

Table 4: Layered intellectual property framework

No Patent Strategy: The Licensor has deliberately chosen not to pursue patent protection for the AMAGI Architectural Class. The AMAGI strategy is open standard plus trademark plus certification, not patent enforcement. This decision reflects the philosophical commitment to making the architecture as widely adoptable as possible while maintaining control through the S-APL contractual framework and the AMAGI certification mark. No patent applications have been filed, and none are planned. The AMAGI Architectural Class will remain patent-free for the entire lifetime of the specification.

Trade Secret Protection: The specific implementation details of the SCC FSM, EGW, and CRTM are maintained as trade secrets of the Licensor. Licensees who receive these details under NDA as part of the compliance verification process are bound by trade secret obligations in addition to the S-APL. This dual protection (contractual + trade secret) ensures that implementation details are protected without relying on patent law, consistent with the open standard strategy.

Independent Implementation Defense: A party that independently develops a hardware-enforced AI safety architecture without reference to the AMAGI specifications, and without receiving AMAGI trade secrets, is not bound by the S-APL. However, such a party may not use the AMAGI trademark, may not claim AMAGI-class compliance, and may not represent its implementation as AMAGI-class. The burden of proving independent development rests on the party claiming it. This License does not restrict the right to independently develop alternative architectures that achieve similar safety properties through different structural mechanisms.

9. Zero-Knowledge Proof Audit Protocol

The Zero-Knowledge Proof (ZKP) Audit Protocol enables a Licensee to demonstrate compliance with the Audit Evidence Clause (Section 7) without revealing the underlying audit records that may contain PII or PHI. The protocol is designed to satisfy three simultaneously held requirements: (1) the Licensor and relevant regulators can verify that the AMAGI invariants are being maintained in operation; (2) the Licensee does not disclose PII or PHI to third parties in violation of GDPR or HIPAA; and (3) the cryptographic integrity of the audit trail is preserved throughout the verification process.

9.1 Protocol Design

The ZKP Audit Protocol operates on the following principles. The Licensee maintains the complete audit trail locally within the SHMF. When the Licensor or a regulator requests compliance verification, the Licensee generates a zero-knowledge proof that demonstrates the following properties without revealing individual audit records:

- **Invariant Compliance Proof:** A zk-SNARK proof that all SCC decisions recorded in the audit trail during the verification period satisfied the TTP predicate (i.e., every activation was preceded by a valid trigger, delay, and policy evaluation).

• **Audit Chain Integrity Proof:** A Merkle proof that the audit trail has not been tampered with, demonstrating that the hash chain from the first record to the last record is intact and that all Amagi-Q signatures are valid.

• **No Kill Switch Activation Proof:** For Tier 2 systems, a proof that the emergency termination provision (Section 10) has not been invoked during the verification period.

• **Policy Compliance Proof:** A proof that all MAG policy evaluations resulted in outcomes consistent with the deployed policy matrix, without revealing the specific policy entries or the data that was evaluated.

9.2 Implementation Status

The ZKP Audit Protocol is a conceptual design at the time of this License publication. The Licensor commits to developing a reference implementation of the ZKP Audit Protocol within 18 months of the effective date of this License. Until the reference implementation is available, the transitional provisions of Section 7.2 (anonymized audit trail excerpts) apply. The ZKP Audit Protocol, when implemented, will be published as an amendment to this License and will not require a new license version.

The ZKP circuit design will target the Groth16 proving system with BN254 curve, providing succinct proofs that can be verified in under 10 milliseconds on commodity hardware. The trusted setup ceremony will be conducted using a multi-party computation protocol with at least 100 participants from independent organizations to ensure the toxic waste is destroyed. The verification key will be published as part of the AMAGI Open Specification.

9.3 Regulatory Acceptance

The Licensor will engage with the relevant data protection authorities (the Information Commissioner's Office in the UK, the European Data Protection Board, and the HHS Office for Civil Rights in the US) to obtain formal recognition that the ZKP Audit Protocol satisfies the audit evidence requirements of the S-APL while maintaining compliance with GDPR and HIPAA. The Licensor will also seek recognition from CC EAL2+ and IEC 61508 SIL2 certification bodies that ZKP-based audit evidence is an acceptable alternative to full audit trail disclosure.

10. Emergency Termination Provision

10.1 Purpose and Scope

The Emergency Termination Provision (colloquially, the "Kill Switch") allows the Licensor to remotely disable an AMAGI-class implementation under strictly defined circumstances where the implementation has been confirmed to pose an imminent threat to human safety. This provision exists as a last-resort safety mechanism and is not intended for routine enforcement of license compliance or commercial disputes.

The Emergency Termination Provision operates through a hardware-enforced mechanism: the SCC contains a dedicated termination latch that, when asserted, forces the system into a permanent CUTOFF state. The termination latch can be asserted only by a cryptographically authenticated command signed by the Licensor's emergency key (ML-DSA-44, stored in a hardware security module with multi-party access control). The termination latch cannot be reset by software; recovery requires physical replacement of the OTP module.

10.2 Tier 2 Emergency Termination

For Tier 2 systems, the Emergency Termination Provision is available subject to the following conditions:

- **Imminent Threat:** The Licensor must have received a verified report from a certified safety assessor, a regulatory body, or the Licensee itself that the AMAGI-class implementation poses an imminent threat to human safety that cannot be mitigated through the normal TTP intervention mechanisms.
- **Independent Verification:** The threat must be independently verified by at least two of the following: a certified safety assessor, a regulatory body with jurisdiction, or an accredited independent assessor designated by the Licensor.
- **Due Process:** The Licensee must be given 72 hours notice before termination, during which the Licensee may contest the termination by providing evidence that the threat has been mitigated or does not exist. The 72-hour notice period may be shortened to 4 hours only in cases where the imminent threat involves risk of death or serious injury.
- **Post-Termination Review:** Within 30 days of any emergency termination, the Licensor must conduct a full review of the circumstances and publish a summary report (redacted of PII/PHI) to the AMAGI governance board.

10.3 Tier 1 Carve-out

The Emergency Termination Provision does not apply to Tier 1 systems. This carve-out is necessary because the presence of a remote termination mechanism is fundamentally incompatible with the CC EAL2+ augmented certification that Tier 1 systems target. Specifically, the Common Criteria (ISO/IEC 15408) requires that the TOE (Target of Evaluation) has no external mechanism that can override its security functions; a remote kill switch would constitute such a mechanism and would prevent the TOE from achieving EAL2+ augmented certification.

The Tier 1 carve-out is enforced at the hardware level: the termination latch in Tier 1 systems is permanently disabled (fused off) during manufacture. This means that Tier 1 systems are not susceptible to remote termination by the Licensor, by the Licensee, or by any third party. The trade-off is that Tier 1 systems must rely entirely on their own TTP intervention mechanisms and the hardware kill switch (activated by local safety sensors) for safety response. This is consistent with the Tier 1 design philosophy: maximum assurance through minimum external dependency.

Implication for Licensees: A Licensee selecting Tier 1 accepts that the Licensor cannot remotely intervene in the operation of the system, even in an emergency. The Licensee assumes full responsibility for the safety of the deployment and must implement its own emergency response procedures in accordance with the applicable safety standards (IEC 61508, ISO 26262, DO-178C, IEC 62304 as applicable).

11. Succession and Key Person Risk

11.1 Key Person Risk

The AMAGI Architectural Class was created by a single architect (the Key Person, Alexey M. Burlai), who currently holds the Golden Share in AGIAM Technologies Ltd and serves as the sole authority for license issuance, certification decisions, emergency termination (Tier 2), and the custody of trade secrets and emergency cryptographic keys. This concentration of authority in a single individual creates a Key Person Risk: if the Key Person becomes incapacitated, deceased, or otherwise unable to perform their duties, the AMAGI governance and licensing infrastructure could become inoperable, leaving existing Licensees without access to compliance verification, license renewal, or emergency termination services.

11.2 Succession Protocol

To mitigate Key Person Risk, the following Succession Protocol is established:

- **Successor Designation:** The Key Person shall designate a Successor in writing, filed with the board of AGIAM Technologies Ltd and with a qualified escrow agent. The Successor must be a person with demonstrated expertise in hardware security architecture, formal verification, and AI safety, and must be approved by a two-thirds majority of the AMAGI governance board (when constituted).
- **Emergency Key Escrow:** The Licensor's emergency cryptographic key (used for Tier 2 emergency termination) shall be split using Shamir's Secret Sharing (k-of-n threshold, k=3, n=5) and the shares shall be escrowed with independent trustees: the Key Person, the designated Successor, the AMAGI governance board chair, and two independent trustees from accredited certification bodies. This ensures that emergency termination can be invoked even if the Key Person is unavailable, while preventing any single party from invoking it unilaterally.
- **Golden Share Transfer:** Upon the death or permanent incapacity of the Key Person, the Golden Share shall transfer to the designated Successor in accordance with the articles of association of AGIAM Technologies Ltd and the Key Person's last will and testament. The transfer shall be effective upon receipt by the board of a certified copy of the death certificate or a medical certificate of permanent incapacity. The Successor assumes all rights and obligations of the Key Person under this License, including the emergency termination authority and the custody of trade secrets.

- **Interim Governance:** If the Key Person becomes temporarily incapacitated (expected recovery within 90 days), the AMAGI governance board shall appoint an interim authority to perform the Key Person's duties. The interim authority may not modify the S-APL, issue new licenses on terms different from the current standard terms, or invoke emergency termination without the k-of-n threshold described above.
- **Continuity of Obligations:** All licenses issued under the S-APL remain in full force and effect regardless of any succession event. The obligations of existing Licensees are not affected by a change in the Key Person or the holder of the Golden Share. The Licensor (AGIAM Technologies Ltd) is the contracting party, not the Key Person individually, and the License is a contract with the company, not with the individual.

11.3 Review and Update

The Succession Protocol shall be reviewed annually by the AMAGI governance board and updated as necessary to reflect changes in the organizational structure, the availability of qualified successors, and the evolving regulatory landscape. The Key Person shall certify annually that a valid Successor designation and current emergency key escrow arrangements are in place. Failure to certify for two consecutive years triggers an automatic governance board review and the potential appointment of an interim authority.

12. Dual-Use and Export Control

12.1 Dual-Use Classification

The AMAGI Architectural Class is a dual-use technology: it is designed for safety-critical AI applications in civilian domains (healthcare, automotive, aerospace, finance) but its hardware-enforced non-bypassability properties could also be applied in military, intelligence, or weapons systems. The Wassenaar Arrangement on Export Controls for Conventional Arms and Dual-Use Goods and Technologies classifies certain cryptographic and security technologies under Category 5 (Information Security) of the Dual-Use Goods List. The AMAGI Architectural Class, by virtue of its hardware-enforced access control, cryptographic audit, and tamper-resistant design, may fall within the scope of these controls.

The Licensor classifies the AMAGI specifications as dual-use technology under the Wassenaar Arrangement and the applicable national export control regulations (including the EU Dual-Use Regulation 2021/821, the US Export Administration Regulations (EAR), and the UK Strategic Export Control Order 2008). The S-APL does not grant any exemption from applicable export control laws, and the Licensee is solely responsible for obtaining all necessary export licenses, permits, and authorizations before exporting, re-exporting, or transferring an AMAGI-class implementation or the related technical data.

12.2 Liability Disclaimer

The Licensor makes no representation or warranty regarding the export control classification of any specific AMAGI-class implementation, as the classification depends on the implementation details, the destination country, the end user, and the end use — all of which are within the knowledge and control of the Licensee, not the Licensor. The Licensee assumes all liability for export control violations arising from the implementation, deployment, or transfer of an AMAGI-class system.

The Licensor shall not be liable for any loss, damage, or penalty arising from: (a) the Licensee's failure to obtain necessary export licenses; (b) the Licensee's deployment of an AMAGI-class system in a jurisdiction or for an end use that is prohibited by applicable export control laws; (c) the Licensee's transfer of AMAGI technical data to a sanctioned party or entity; or (d) any regulatory action taken against the Licensee as a result of export control violations.

12.3 Military and Weapons Exclusion

The S-APL expressly prohibits the deployment of AMAGI-class systems in weapons systems, autonomous lethal systems, or any military application where the primary purpose is to cause harm to persons. This prohibition is a fundamental term of the License and its violation constitutes an immediate and automatic termination of the License without notice. The Licensor reserves the right to report any such violation to relevant regulatory and enforcement authorities.

This exclusion does not apply to defensive military applications (e.g., missile defense systems, cybersecurity of military infrastructure, or medical AI in military hospitals) provided that the primary purpose of the system is protection rather than harm. The determination of whether a specific deployment falls within the exclusion is made by the Licensor in consultation with the AMAGI governance board and relevant regulatory authorities.

13. Compliance Verification Requirements

Licensees under the S-APL are required to submit compliance evidence at three stages:

13.1 Design Compliance

Design Compliance verifies that the HDL implementation of the TCB satisfies the 500-line complexity bound and that the SPIN model checking results confirm all safety and liveness properties. The Licensee must submit: (a) the HDL source code of the SCC core; (b) the SPIN/Promela model and verification results; (c) the HDL line count certification; and (d) the trust boundary specification showing the Normal World / Secure World partition.

Netlist-to-RTL Equivalence (Tier 1 Requirement): For Tier 1 / Hard Assurance systems, the Licensee must additionally submit a formal Netlist-to-RTL equivalence check performed after logic synthesis. This requirement protects INV-002 (Irreducible TCB) against post-synthesis violations: CAD optimizer transformations (retiming, resource sharing, state encoding changes) may increase the effective TCB beyond the 500-line HDL limit or introduce logic that violates the TTP predicate. The equivalence check must demonstrate that the post-synthesis netlist is functionally equivalent to the pre-synthesis RTL within the TCB boundary, using a formal equivalence checking tool (e.g., Cadence Conformal LEC, Synopsys Formality). Any discrepancy between the RTL and netlist representations of the TCB constitutes a Design Compliance failure and must be resolved before the implementation may proceed to Deployment Compliance (Section 13.2).

> **Rationale:** The Netlist-to-RTL equivalence requirement addresses a supply chain threat identified in the cross-document consistency audit: if a Tier 1 licensee receives RTL (VHDL/Verilog) under S-APL and performs their own synthesis, the CAD optimizer may violate INV-002 by expanding the TCB beyond the 500-line limit or by introducing logic that undermines the TTP predicate. The equivalence check ensures that the structural guarantees verified at the RTL level are preserved through the synthesis flow to the final gate-level netlist.

13.2 Deployment Compliance

Deployment Compliance verifies that the physical implementation maintains the trust boundaries defined in the AMAGI Framework, that the MAG has been correctly programmed into eFuse/OTP, and that the selected trust tier (Tier 1 or Tier 2) is documented and enforced. The Licensee must submit: (a) the OTP programming verification report; (b) the trust tier declaration; (c) the EGW fuse configuration for Tier 1 systems (termination latch disabled); and (d) the physical security assessment of the deployment environment.

13.3 Operational Compliance

Operational Compliance requires periodic audit trail review and, for Tier 2 systems, verification that all policy updates were performed in accordance with the Dual-Tier Trust Model procedures. The Licensee must submit: (a) the annual audit trail integrity verification report (full trail or ZKP proof per Section 9); (b) the OTA update log for Tier 2 systems; (c) the incident report for any safety interventions during the reporting period; and (d) the certification renewal assessment.

Compliance verification is performed by AGIAM Technologies Ltd or by an accredited independent assessor designated by AGIAM. The cost of compliance verification is borne by the Licensee.

14. Term, Termination, and Remedies

14.1 Term

This License is granted for the operational lifetime of the AMAGI-class implementation, unless earlier terminated in accordance with this Section. The License is not transferable except with the prior written consent of the Licensor, which shall not be unreasonably withheld.

14.2 Termination for Breach

The Licensor may terminate this License upon 30 days written notice if the Licensee materially breaches any term of this License and fails to cure the breach within the 30-day notice period. Material breach includes, without limitation: (a) deployment of a Derivative Architecture as AMAGI-class; (b) failure to comply with the compliance verification requirements; (c) violation of the Military and Weapons Exclusion (Section 12.3); and (d) unauthorized transfer of the License.

14.3 Effect of Termination

Upon termination, the Licensee must: (a) cease all commercial use of the AMAGI trademark and certification marks; (b) cease representing the implementation as AMAGI-class; (c) destroy or return all AMAGI trade secret materials in its possession; and (d) provide the Licensor with a certified statement of compliance with this Section. Termination does not affect the Licensee's right to continue operating the implementation under a different name and without AMAGI-class claims, provided that the implementation does not constitute a Derivative Architecture that misrepresents itself.

14.4 Remedies

The Licensor's remedies for breach of this License include: (a) injunctive relief to prevent further breach; (b) damages for losses caused by the breach; (c) specific performance of compliance verification requirements; and (d) public disclosure that the Licensee's implementation is not AMAGI-class. The Licensor's right to public disclosure is a specific remedy for the reputational harm caused by misrepresentation of Derivative Architectures as AMAGI-class.

15. Disclaimers and Limitation of Liability

The Licensed Material is provided "as-is" without warranty of any kind, express or implied, including but not limited to the warranties of merchantability, fitness for a particular purpose, and non-infringement. In no event shall the Licensor be liable for any claim, damages, or other liability, whether in an action of contract, tort, or otherwise, arising from, out of, or in connection with the Licensed Material or the use or other dealings in the Licensed Material.

The Licensor does not warrant that the AMAGI Architectural Class will achieve any specific certification outcome (CC EAL2+, IEC 61508 SIL2, or otherwise), that the TTP predicate will prevent all safety incidents, or that the implementation will be free from defects. The Structural Guarantee under this License is a guarantee of architectural compliance, not a guarantee of operational safety. The Licensee is solely responsible for the safety of its deployment and must conduct its own safety assessment in accordance with applicable standards.

16. Governing Law and Dispute Resolution

This License is governed by the laws of England and Wales. Any dispute arising out of or in connection with this License, including any question regarding its existence, validity, or termination, shall be referred to and finally resolved by arbitration under the LCIA (London Court of International Arbitration) Rules, which Rules are deemed to be incorporated by reference into this clause. The number of arbitrators shall be three. The seat, or legal place, of arbitration shall be London, England. The language of the arbitration shall be English. The governing law of the arbitration shall be the laws of England and Wales.

Notwithstanding the foregoing, the Licensor may seek injunctive relief in any court of competent jurisdiction to prevent irreparable harm arising from a breach of the Derivative Architecture Clause

(Section 5) or the Military and Weapons Exclusion (Section 12.3).

Appendix A: Derivative Architecture Determination Test

The following decision tree may be used to determine whether an implementation is AMAGI-class or a Derivative Architecture. This test is not exhaustive; the Licensor retains the final determination authority.

Step	Question	If Yes	If No
1	Does the implementation enforce all six invariants?	Proceed to Step 2	Derivative Architecture
2	Is the TTP predicate enforced in hardware (not software)?	Proceed to Step 3	Derivative Architecture (software-only)
3	Is the SCC TCB no more than 500 lines of HDL?	Proceed to Step 4	Derivative Architecture (TCB expansion)
4	Is the MAG stored in OTP/eFuse (immutable or PQC-OTA)?	Proceed to Step 5	Derivative Architecture (policy mutability)
5	Does the implementation use the AMAGI structural pattern (Trigger-Delay-Policy-Execute-Audit)?	AMAGI-class; S-APL license required	Independent architecture; S-APL does not apply

Table 6: Derivative architecture determination test

Appendix B: ZKP Audit Protocol — Conceptual Design

This appendix provides a high-level conceptual design of the Zero-Knowledge Proof Audit Protocol. The detailed cryptographic specification will be published separately as part of the AMAGI Open Specification upon completion of the reference implementation.

B.1 Cryptographic Primitives

The ZKP Audit Protocol uses the following cryptographic primitives: (1) zk-SNARK (Groth16) with BN254 pairing-friendly curve for proof generation and verification; (2) Merkle tree (SHA-256) for audit trail commitment; (3) ML-DSA-44 (FIPS 204) for signature verification within the ZKP circuit; and (4) Shamir's Secret Sharing for emergency key escrow. These primitives are selected for their compatibility with the existing AMAGI cryptographic infrastructure and their well-established security properties.

B.2 Proof Generation Workflow

Step 1: The Licensee extracts the relevant invariant compliance predicates from the audit trail for the verification period (typically one year).

Step 2: The Licensee constructs a Merkle tree of the audit trail and computes the root commitment.

Step 3: The Licensee generates a zk-SNARK proof that: (a) the Merkle root commits to a valid audit trail; (b) all TTP predicate evaluations in the committed trail resulted in correct outcomes; (c) all Amagi-Q signatures are valid; and (d) no emergency termination was invoked (for Tier 2).

Step 4: The Licensee submits the zk-SNARK proof and the Merkle root commitment to the Licensor or regulator.

Step 5: The verifier (Licensor or regulator) verifies the zk-SNARK proof against the published verification key. Verification takes less than 10 milliseconds and reveals no information about individual audit records.

B.3 Security Properties

The ZKP Audit Protocol provides the following security properties: (1) **Completeness:** If the Licensee is honest and the audit trail is valid, the proof will always verify. (2) **Soundness:** If the Licensee is dishonest and the audit trail contains violations, the Licensee cannot generate a valid proof except with negligible probability (2^{-128} under the BN254 curve). (3) **Zero-Knowledge:** The proof reveals no information about individual audit records beyond the statements being proven. (4) **Succinctness:** The proof size is constant (approximately 200 bytes for Groth16) regardless of the number of audit records. (5) **Non-Repudiation:** The Merkle root commitment binds the Licensee to a specific audit trail; the Licensee cannot later claim a different trail was used.

Appendix C: Succession Protocol Summary

Event	Trigger	Successor Activation	Emergency Key Access
Death of Key Person	Certified death certificate received by board	Immediate upon board confirmation	3-of-5 Shamir threshold (Successor + board chair + 1 trustee)
Permanent Incapacity	Medical certificate of permanent incapacity	30 days after board confirmation	3-of-5 Shamir threshold
Temporary Incapacity	Expected recovery within 90 days	Interim authority appointed by board (no Successor activation)	3-of-5 Shamir threshold (interim authority replaces Key Person share)
Voluntary Resignation	Written notice to board	90-day transition period	Key Person retains share during transition; Successor receives share on activation

Table 7: Succession protocol event summary